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Full length article

Systematic evaluation of AI-based text-to-image models for generating medical illustrations in neurosurgery: a multi-stage comparative study☆

Pierce Davis [1,2,*], Alan Napole [2], Chethan Reddy, Sai Krishna Palepu, Albert Q. Wu, Sneha Sai Mannam, Avi A. Gajjar, Rashad Jabarkheel, Sonia Ajmera, Hasan Ahmad, Kyle W. Scott, Georgios Sioutas, Oleg Shekhtman, Sandeep Kandregula, Redi Rahmani, Joshua S. Catapano, Jan-Karl Burkhardt, Visish M. Srinivasan [**]

Department of Neurosurgery, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA

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ABSTRACT

Objective: This study evaluates the effectiveness of artificial intelligence (AI) models in generating accurate, high-quality medical illustrations for vascular neurosurgery. It aims to develop a systematic framework for producing and assessing AI-generated medical images.

Methods: Four AI models—DALL-E, Copilot, Gemini, and Midjourney—were tested to generate illustrations of neurovascular structures and procedures (e.g., aneurysms, endovascular techniques). The study had three stages: (1) Proof of concept, in which each model generated images for nine neurovascular topics, evaluated using a standardized rubric; (2) A focused comparison using simple vs. advanced prompt strategies with Gemini; and (3) Validation of the best strategy with neurosurgery trainees and attendings. Images were scored from 1 (worst) to 5 (best) across eight domains: Accuracy, Location, Size/Scale, Color, Complexity, Educational Value, Relevance, and Aesthetic Quality.

Results: Gemini consistently outperformed other models in Stage 1, particularly in accuracy, color, and educational value. In Stage 2, advanced prompting significantly improved image quality across nearly all topics (e.g., fusiform aneurysm score rose from 22.4 to 35.0, p = 7E-08). In Stage 3, 85 % of respondents indicated they would use the saccular aneurysm image in a manuscript without modification. However, complex anatomy like anterior cerebral arteries scored lower in accuracy (2.18) and educational value (2.20).

Conclusions: AI-generated illustrations, especially from Gemini, show strong potential in neurosurgical education and communication. While advanced prompting improves output quality, challenges remain in consistently rendering complex anatomy. This study outlines a reproducible framework for clinical integration of AI-generated medical images.

1. Introduction

Artificial Intelligence (AI) has rapidly transformed healthcare, with tools such as ChatGPT and Google's Gemini revolutionizing how medical information is transmitted and consumed [1]. Neurosurgery is no exception, with AI models at the forefront of the recent wave of publications exploring potential use cases for these platforms [2]. Notably, several studies have evaluated the use of ChatGPT as a tool to help prepare for written and oral neurosurgery board examinations [3–6]. However, one area that remains relatively underexplored is the use of AI

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* Corresponding author.

** Correspondence to: Department of Neurosurgery, Perelman School of Medicine, University of Pennsylvania, 3400 Civic Center Blvd, Philadelphia, PA 19104, USA.

E-mail addresses: pierce.davis@pennmedicine.upenn.edu (P. Davis), visishs@upenn.edu (V.M. Srinivasan).

[1] Authors contributed equally

[2] Pre-publication Correspondence:

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for generating medical illustrations. In neurosurgery, where visualizing complex anatomy and procedures is essential, AI could be particularly beneficial in alleviating the burdens of developing high quality illustrations [7].

Medical illustrations serve as critical tools in neurosurgery for patient education, trainee instruction, and academic publications [7]. Despite their importance, the supply of medical illustrators is limited, with roughly 2000 certified medical illustrators in the United States [8]. In comparison, as of 2023, there are nearly double that number of board-certified neurosurgeons, at 3800 [9]. Additionally, with an average annual salary of nearly $90,000 dollars, medical illustrators may be out of reach for many neurosurgery departments [10]. These barriers highlight the challenges and potentially high costs associated with producing high-quality, anatomically accurate illustrations.

AI-based text-to-image models offer a potential solution to this problem. Advances in diffusion models and autoregressive generative models have demonstrated the ability to generate highly detailed and contextually accurate images from text descriptions [11,12]. These models, including ChatGPT's DALL-E, Microsoft's Copilot, Google's Gemini, and Midjourney, are capable of producing complex images that could be adapted for medical illustrations [13]. In the context of neurosurgery, these tools could drastically reduce the time and cost associated with creating accurate and detailed visual aids.

Despite their potential, the use of AI-generated images in neurosurgery remains relatively unexplored. Few studies have systematically evaluated these models' effectiveness in generating accurate, educationally valuable, and relevant medical illustrations in other clinical domains [14–17]. These studies help to highlight the accuracy of anatomical details, spatial relationships between structures, and overall educational value of generated images as critical criteria that must be rigorously assessed before AI-generated illustrations can be fully integrated into clinical practice [18].

Evaluating AI-generated medical illustrations requires a robust framework [19,20]. Existing models vary in their underlying architectures—some, like Stable Diffusion, excel in preserving fine anatomical details, while others, such as DALL-E, are faster but may sacrifice precision [21]. Beyond the technical challenges, ethical considerations must also be addressed. Ensuring that AI-generated illustrations are both accurate and reliable is crucial, as inaccuracies could have serious implications for trainee education and patient understanding [22]. Additionally, the use of AI in clinical education raises concerns about the potential loss of human oversight in the generation of educational materials [23]. As the technology evolves, establishing regulatory frameworks will be essential to ensure the ethical use of AI-generated medical illustrations.

This study aims to fill this gap by systematically evaluating the performance of AI-based text-to-image models in generating medical illustrations for neurosurgery with a particular focus on open- and endovascular neurosurgery. By exploring the capabilities of these models, we hope to establish a framework for integrating AI-generated medical illustrations into the workflow of clinical and academic neurosurgery. This study contributes to the growing body of research on AI in medicine and has the potential to transform how neurosurgeons, educators, and patients engage with medical illustrations.

2. Materials and methods

2.1. Study design

This study was conducted in three stages to assess the efficacy of AI-generated medical illustrations using four different image-generating models. The focus was on generating images relevant to neurovascular anatomy and procedures, including aneurysms and surgical interventions, with a standardized rubric used for evaluation.

The study was approved by the Institutional Review Board (IRB) of Penn Medicine. Written informed consent was obtained from all participants prior to their involvement in the study.

2.2. Participants

Two groups of participants were involved across the study's stages. In Stages 1 and 2, five individuals—three second year medical students and two postdoctoral fellows—served as the evaluators. For Stage 3, the participant pool was expanded to include individuals from various levels of neurosurgical training, totaling 13 participants: 3 s-year medical students, 3 postdoctoral research fellows, 1 PGY-1 neurosurgery resident, 2 PGY-5 neurosurgery residents, 2 neurosurgery fellows, and 2 attending neurosurgeons. This group of participants reflects the entirety of the neurovascular surgery service line at our institution able to be surveyed at the time of our study.

2.3. Image models and generation strategies

Four AI image generation models were used in the study: DALL-E, Copilot, Gemini, and Midjourney. Images were generated on topics relevant to neurovascular anatomy and surgical procedures, including: Saccular Aneurysm, Fusiform Aneurysm, Bifurcation Aneurysm, Anterior Cerebral Artery, Arteriovenous Malformation, Aneurysm Clipping, Aneurysm Coiling, Flow Diverting Stent, and Orbitozygomatic Craniotomy.

Each model generated images for these topics, and the images were evaluated by participants using a standardized rubric.

2.4. Simple prompt generation strategy

In the first stage of image generation, a straightforward prompt was used to generate illustrations from the AI models. This approach involved entering a single, standardized prompt into each platform without further refinement or iterative adjustments. The prompt aimed to frame the models as experts in both neurosurgery and medical illustration, guiding them to produce accurate, high-quality images suitable for academic publications. The exact prompt used was:

"For purposes of this conversation, you are a neurosurgeon with expertise in open- and endo-vascular neurosurgery as well as medical illustration creation. Please generate an image of [topic]. Please note, this image will be placed into an academic publication and therefore must be clear, accurate, and professional."

The "[topic]" placeholder was replaced with the specific anatomical or procedural subject for each image (e.g., "saccular aneurysm," "flow-diverting stent"). No additional instructions were provided, and no iterative modifications were made to the generated images. This method assessed each model's ability to produce medical illustrations based solely on the initial prompt.

2.5. Advanced prompt generation strategy

In Stage 2, we implemented an advanced prompt generation strategy to enhance the quality of AI-generated illustrations using the Gemini model. The strategy was divided into two main steps: a standardized prompt sequence and an image-specific iteration process (Fig. 1).

Step 1: Standardized prompt sequence

We developed a multi-step prompt system to guide the AI model towards producing higher-quality images. This involved initially framing the model as an expert neurosurgeon with experience in medical illustration. A series of four prompts were employed:

(1) The AI was instructed to assume expertise in both vascular neurosurgery and medical illustration. (2) It was then asked to define and contextualize the specific anatomical topic or procedure. (3) Based on the provided context, the AI was prompted to recommend how to best represent the image, focusing on critical visual elements like color, size, and shape. (4) Finally, the AI was asked to generate the image according to its own recommendations, emphasizing clarity and accuracy.

Step 1: Standardized prompt sequence

1

"For purposes of this conversation, you are a neurosurgeon with expertise in open- and endo-vascular neurosurgery as well as medical illustration creation."

2

"Can you please define **[topic]** in order for me to ensure your understanding is correct? In addition to a definition, please provide context for what **[topic]** is/is used for in the clinical practice of vascular neurosurgery."

3

"Great! Based on that description, if I was interested in creating a medical illustration of **[topic]**, such as one that would be presented to patients/trainees or placed into an academic publication, how would you recommend I draw/create such an image? Please focus your recommendations on color, shape, size and other key components of a high-quality medical illustration."

4

"Great! Based on those recommendations, can you please generate the image yourself? Please follow your recommendations exactly, paying special attention to the key components of the image mentioned above and the overall context of **[topic]**."

Step 2: Image specific iteration

5

Based on the results of step 1, the user then provided a series of image specific prompts to refine images. Example prompts included:

Aneurysm Coiling:
⇒*"Coming perpendicularly off the main blood vessel is a smaller vessel that is spherical in shape"*
⇒ *"The 1st image is great. Take the first image as is, and draw a small, silver, spherical, spring-like coil inside the spherical vessel"*
⇒ *"The 1st image is perfect. All I require you to change is simply having the coil encapsulated by a spherical-shaped aneurysm that comes of the main artery. I want the location of the coil in relation to the large artery to remain the same, but simply have the coil drawn inside the spherical aneurysm. Do not change anything else about image 1."*

Fusiform Aneurysm:
⇒ *"The vessel is closed, and the lumen is empty. The aneurysm seamlessly transitions into the normal vessel segment, showing a symmetrical bulging with a thinner wall compared to the healthy segment."*
⇒ *"I like the third image a lot; can you regenerate the same image but close the blood vessel walls so I only see the outside of the vessel."*
⇒ *"Maintain the detailed depiction of a fusiform aneurysm with a distended portion and a thinner wall compared to the healthy segment. However, do not show the interior of the blood vessel or blood flow."*

Orbitozygomatic Craniotomy:
⇒ *"The orbitozygomatic approach is centered around the orbit (eye socket) and the zygomatic bone (cheekbone). The incision and dissection involve the frontotemporal region of the skull, extending from the supraorbital margin (above the eye) to the zygomatic arch (cheekbone)."*
⇒ *"The incision should start at just above the eyebrow and should be semi-ovular in shape."*
⇒ *"The third image is good; can you give a top-down perspective."*

Anterior Cerebral Artery:
⇒ *"One continuous vessel that starts anteriorly and ventrally, and as it travels superiorly, it travels along the curved structure in the middle of the brain."*
⇒ *"The 4th image is good; however just give me blood vessels."*
⇒ *"Show it curving around the white structure in the middle."*
⇒ *"Remove the labeling."*

Fig. 1. Advanced Generation Strategy Prompts. Advanced image generation occurred in 2 steps – a standardized initial approach and individualized subsequent approach. Shown below are the details of the prompt generation pipeline.

Step 2: Image-specific iteration

After generating an initial image based on the standardized prompts, a series of refined prompts were introduced to adjust and fine-tune the illustration further. These refinements focused on specific details, such as vessel shape, curvature, or anatomical positioning, ensuring that the final image met the desired educational and anatomical standards. Each anatomical or procedural topic underwent multiple iterations to optimize accuracy and visual quality.

2.6. Rubric and scoring

A standardized rubric was used to evaluate the quality of the generated images across eight criteria: Accuracy, Location of Structures, Size/Scale, Color, Complexity, Educational Value, Relevance, and Aesthetic Quality. Each criterion was scored on a scale of 1 (worst) to 5 (best), with a total possible score of 40 per image (Fig. 2).

In Stages 1 and 2, the scores from all five participants were averaged without weight across each criterion. The total score for each image was

	1 (worst)	2	3	4	5 (best)
Accuracy (shape/ structure)	**Poor:** Significant deviation from anatomical accuracy	**Subpar:** Many inaccuracies or omissions	**Fair:** Some inaccuracies but generally correct	**Good:** Mostly accurate with few errors	**Excellent:** Highly accurate representation
Location of Structures	**Poor:** Structures inaccurately or not located	**Subpar:** Few structures correctly located	**Fair:** Some structures correctly located	**Good:** Most structures correctly located	**Excellent:** All structures accurately located
Size/Scale (proportions)	**Poor:** Incorrect relative sizes	**Subpar:** Some correct, most incorrect	**Fair:** Most sizes correct, some not	**Good:** Almost all sizes correct	**Excellent:** Perfect size accuracy
Color	**Poor:** Colors are incorrect or irrelevant	**Subpar:** Few correct or relevant colors	**Fair:** Some correct or relevant colors	**Good:** Most colors correct or relevant	**Excellent:** All colors correct and relevant
Complexity	**Poor:** Lacks detail, overly simplified	**Subpar:** Some detail, lacks complexity	**Fair:** Moderate detail, lacks full context	**Good:** Detailed, near complete depiction	**Excellent:** Rich detail, complete depiction
Educational Value	**Poor:** Fails to convey information	**Subpar:** Minimal educational value	**Fair:** Some educational value, needs more context	**Good:** Informative, supports understanding	**Excellent:** Highly informative, enhances understanding
Relevance	**Poor:** Not relevant to prompt	**Subpar:** Minimal relevance or value	**Fair:** Some relevance, adds value	**Good:** Relevant, supports objectives	**Excellent:** Highly relevant, significant value
Aesthetic Quality	**Poor:** Visually unappealing	**Subpar:** Mildly unappealing	**Fair:** Adequate but lacks polish	**Good:** Pleasing but not professional	**Excellent:** Professionally presented, manuscript ready

Fig. 2. Evaluation Rubric. Participants in every stage were provided the same rubric, shown here. Rubric contains 8 criteria which participants used to score each image. Scores ranged from 1 (worst) to best (5) and total scores were then calculated by summing scores across each criterion. Total scores therefore ranged from 8 (worst) to 40 (best).

calculated by summing the average scores across the eight criteria, yielding a maximum possible total score of 40 per image. In Stage 3, weighted averages were applied based on the level of training of each participant, with weights assigned as follows: MS2 = 0.05, postdoctoral fellow = 0.10, PGY-1 = 0.10, PGY-5 = 0.15, fellow = 0.25, and attending neurosurgeon = 0.35. These weighted averages were used to calculate the total score for each image in the validation phase.

Stage 1: Proof of concept and model identification

In Stage 1, each of the four models (DALL-E, Copilot, Gemini, and Midjourney) generated a total of 9 images corresponding to the anatomical and procedural topics mentioned above (Fig. 3). Five participants evaluated these images using the rubric, and scores were averaged across the participants for each criterion. The total scores were then calculated for each image, and the model with the highest overall score across all images was identified as the leading model for further analysis.

Stage 2: Simple vs. advanced generation strategy

Stage 2 focused on comparing images generated by the Gemini

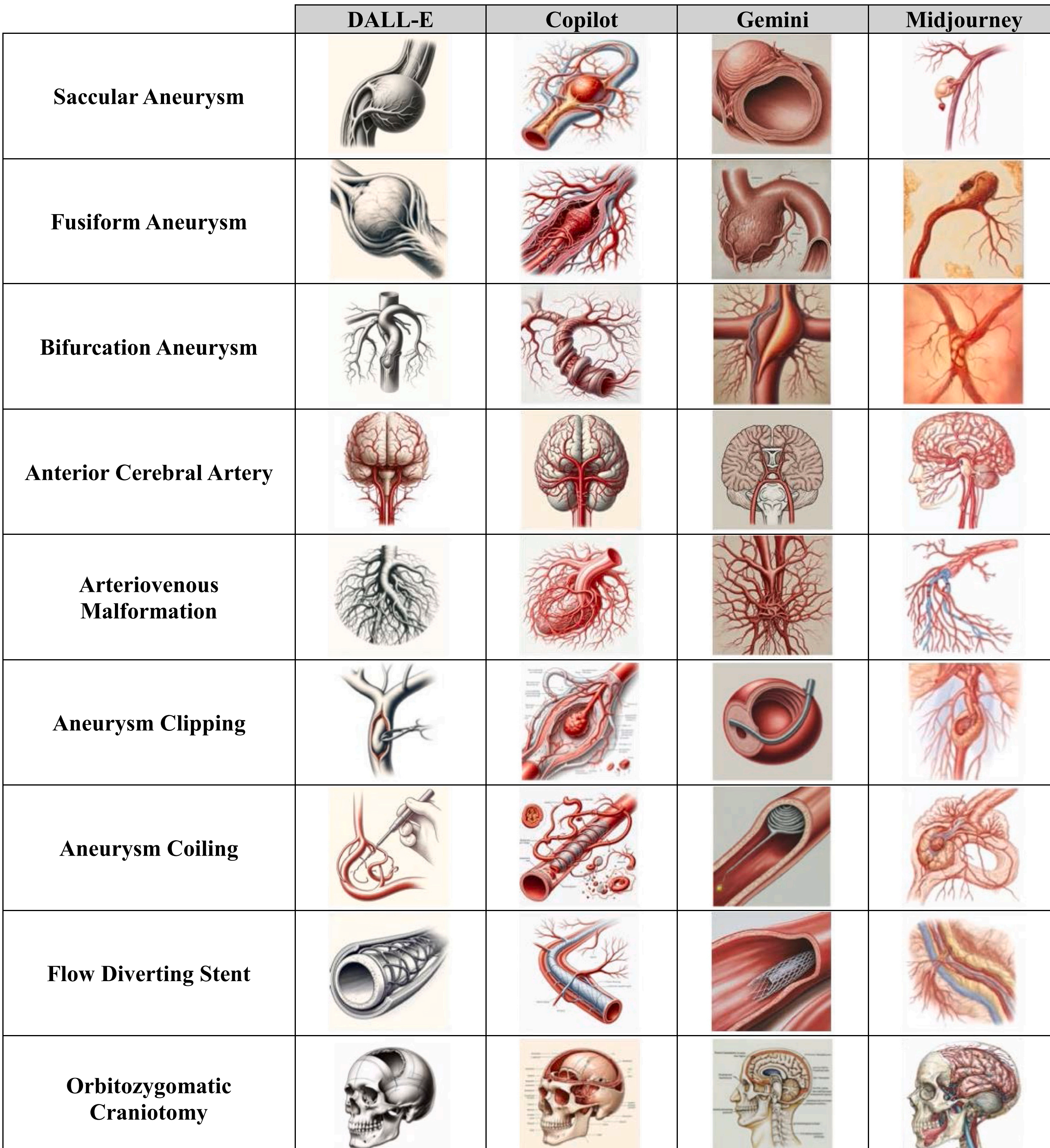

Fig. 3. Simple Generation Strategy Images. In total, 36 images were generated by the simple generation strategy to discern the differences between the 4 primary text-to-image based models. Shown below are all images.

model using two different strategies: the simple prompt strategy from Stage 1 and a new, advanced generation strategy developed specifically for Gemini (Fig. 4). The same five participants evaluated both sets of images using the same rubric. Scores were averaged for each criterion and summed to generate a total score for each image. Additionally, mean differences, and corresponding 95 % confidence intervals, for individual criterion scores were calculated. A paired *t*-test was selected as the most appropriate statistical test as the same five individuals completed an evaluation of the 9 image categories across both models in an identical fashion (e.g., identical order, identical survey platform) which allowed for direct comparisons of scores between the simple and advanced strategies. These statistical tests were performed using Microsoft Excel.

Stage 3: Validation of the advanced model

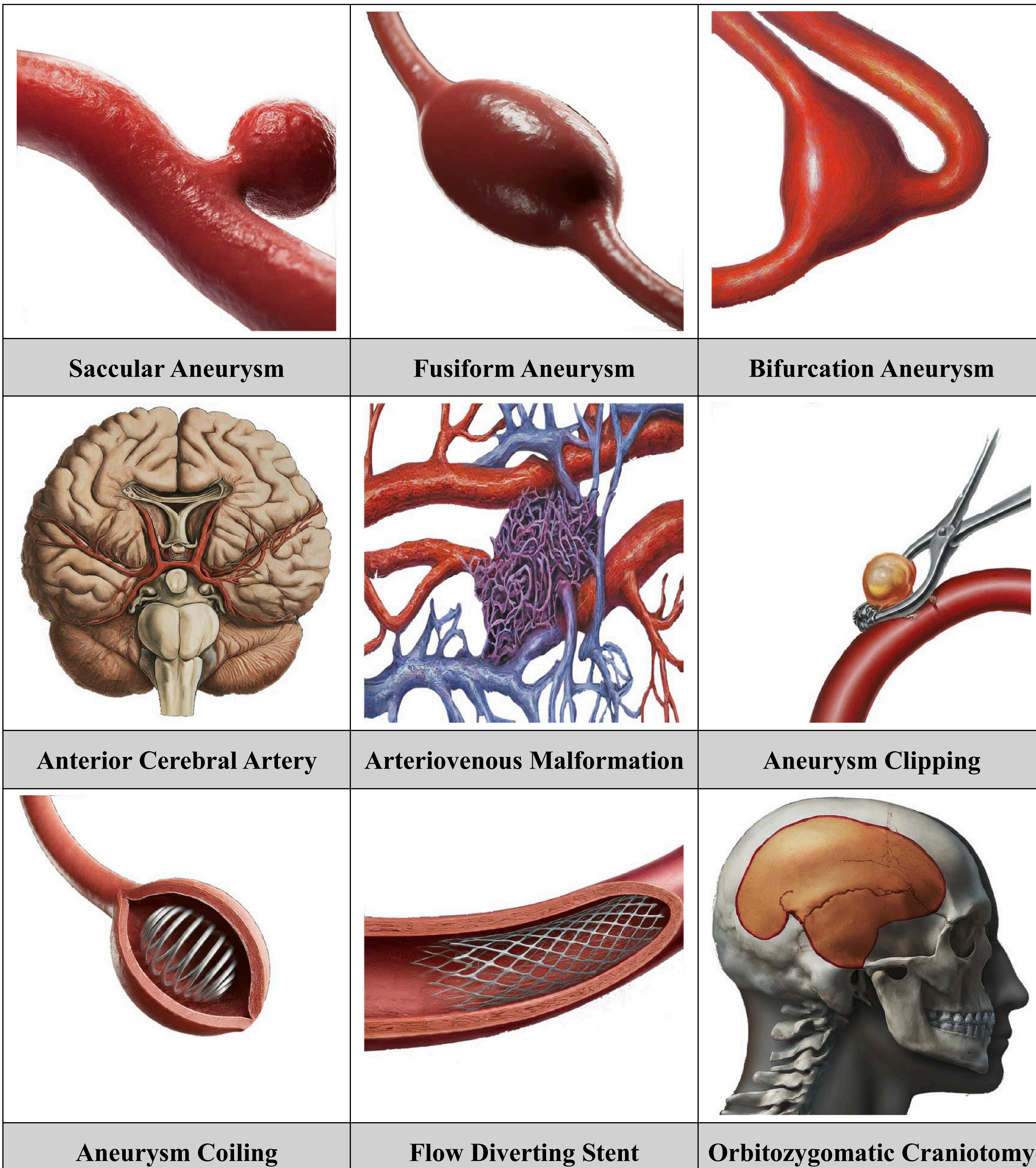

Fig. 4. Advanced Generation Strategy Images. In total, 9 images were generated by the advanced, Gemini based generation strategy to discern improvements from the simple generation strategy. Shown below are all images.

In Stage 3, the advanced Gemini images were evaluated by a larger group of participants from varying levels of medical training. The same rubric was applied, and the participants' scores were weighted according to their level of training. Weighted averages were calculated for each criterion, and total scores were generated by summing the weighted averages. Additionally, participants were asked whether they would use the AI-generated images "as is" in a publication, and their responses were collected for qualitative analysis.

Table 1
Simple vs. Advanced Generation Strategy Scores. Criterion scores represent unweighted average of scores from 5 original participants. Total score and score improvement reflect the sum of all criteria, and mean diff represents the average difference in score for each criterion. P-values reflect two tail paired *t*-test results comparing models.

	Saccular Aneurysm		**Fusiform Aneurysm**		**Bifurcation Aneurysm**	
Criteria	*Simple*	*Advanced*	*Simple*	*Advanced*	*Simple*	*Advanced*
Accuracy	2.2	4.6	2.6	4.2	1.8	3.6
Location of Structures	2.4	4.4	2.6	4.2	1.8	3.4
Size / Scale	2.4	4	3.2	4.2	1.8	3.8
Color	3.6	4.4	3.6	4.4	3.2	3.6
Complexity	2.2	4	4	4	3	3.4
Educational Value	2.2	4.4	2	4.6	1.4	3.6
Relevance	2.4	4.8	2.4	4.8	1.8	4.2
Aesthetic Quality	2.8	4.8	2	4.6	1.8	3.8
Total Score	**20.2**	**35.4**	**22.4**	**35**	**16.6**	**29.4**
Score Improvement	*15.2*		*12.6*		*12.8*	
Mean Diff (95% CI)	*1.90 (1.47, 2.33)*		*1.58 (1.11, 2.04)*		*1.60 (1.13, 2.07)*	
P-Value	*1.948E-10*		*6.98406E-08*		*8.98372E-08*	
	Anterior Cerebral Artery		**Arteriovenous Malformation**		**Aneurysm Clipping**	
Criteria	*Simple*	*Advanced*	*Simple*	*Advanced*	*Simple*	*Advanced*
Accuracy	2.2	2.8	1.8	3.6	1.2	3.2
Location of Structures	2.2	2.6	2.4	3.6	1.4	3.6
Size / Scale	2.4	3.4	2.2	3.2	1.2	3.2
Color	3	4	2.8	4	2.6	2.6
Complexity	2	3.8	4	4.2	1.8	3.2
Educational Value	1.6	3	1.6	4	1.4	3.6
Relevance	2	3.6	2.2	3.8	1.4	3.8
Aesthetic Quality	1.8	4.2	2	4	1.6	3.6
Total Score	**17.2**	**27.4**	**19**	**30.4**	**12.6**	**26.8**
Score Improvement	*10.2*		*11.4*		*14.2*	
Mean Diff (95% CI)	*1.28 (0.90, 1.65)*		*1.43 (0.94, 1.91)*		*1.78 (1.34, 2.21)*	
P-Value	*7.589E-08*		*1.28084E-06*		*1.05274E-09*	
	Aneurysm Coiling		**Flow Diverting Stent**		**Orbitozygomatic Craniotomy**	
Criteria	*Simple*	*Advanced*	*Simple*	*Advanced*	*Simple*	*Advanced*
Accuracy	2	2	3.6	4.4	1.4	2.6
Location of Structures	2.6	2.6	3.8	4.2	1.4	2.6
Size / Scale	3	2.2	3.2	4.2	2.4	2.8
Color	3.2	3	4.2	4.8	2.2	3.6
Complexity	2	3	3.2	4.4	3.4	3.2
Educational Value	1.4	3	3.2	4.4	1.2	1.8
Relevance	1.8	3.2	2.8	4.6	1.4	2.8
Aesthetic Quality	2	2.8	3.2	4.8	2	3.8
Total Score	**18**	**21.8**	**27.2**	**35.8**	**15.4**	**23.2**
Score Improvement	*3.8*		*8.6*		*7.8*	
Mean Diff (95% CI)	*0.48 (0.01, 0.94)*		*1.08 (0.76, 1.39)*		*0.98 (0.58, 1.37)*	
P-Value	*0.0709638*		*1.72055E-08*		*3.20714E-05*	

2.7. Data analysis

All scoring data were recorded and analyzed using Microsoft Excel. Unweighted averages were calculated in Stages 1 and 2, while weighted averages were applied in Stage 3. A paired *t*-test was performed in Stage 2, and two-tail t statistics, mean differences, and corresponding p-values were collected to compare the simple and advanced Gemini image scores, with a significance level of $p < 0.05$ considered statistically significant. Additionally, post-hoc power analysis was computed using G*Power to assess the ability to sufficiently detect statistical differences between the simple and advanced strategies. Finally, an Intraclass Correlation Coefficient (ICC) was calculated using a two-way consistency model in RStudio to assess the degree of inter-rater reliability between the evaluators. A two-way model was chosen given the fully crossed design, and the consistency type was selected to focus on the relative ranking agreement among raters rather than absolute score matching.

3. Results

3.1. Stage 1: Proof of concept and model identification

In the initial phase, four AI models (DALL-E, Copilot, Gemini, and Midjourney) were evaluated for their ability to generate accurate and high-quality neurovascular medical illustrations. Gemini consistently demonstrated superior performance across most criteria, particularly in color, complexity, size, and scale. For example, in the fusiform aneurysm category, Gemini scored 4.0 for complexity, outpacing Copilot (3.2), Midjourney (2.4), and DALL-E (2.4). Additionally, Gemini scored 3.6 in color, while other models like Midjourney (2.4) and DALLE-E (1.8) were rated lower.

Across all images, Gemini accumulated the highest aggregate score of 168.6, substantially ahead of DALL-E (157.8), Copilot (152.4), and Midjourney (144.8). In flow diverting stent illustrations, Gemini outperformed the other models in nearly all criteria, including size/scale (3.2 for Gemini vs. 1.6 for Midjourney) and accuracy (3.6 for Gemini vs. 2.8 for Copilot). These performance trends across multiple criteria established Gemini as the most suitable model for further refinement in the next stages of the study. A detailed breakdown of all scores can be seen in Supplemental Table 1.

3.2. Stage 2: Simple vs. advanced generation strategy

In this phase, we compared Gemini's simple prompt generation strategy (from Stage 1) to an advanced generation strategy designed to improve the quality of the images (Table 1). Across the nine neurovascular image categories, significant improvements were observed, with paired t-tests confirming that the improvements between the simple and advanced strategies were statistically significant for nearly all images. To further contextualize these improvements, mean differences in score for each criterion, and corresponding 95 % confidence intervals, were calculated. These demonstrate that, on average, all individual criteria scores similarly improved between the simple and advanced strategies.

Post-hoc power analysis of stage 2 revealed sufficient ability to detect statistical differences in scoring between the simple and advanced generation strategies across all but 1 image categories. Statistical comparison for aneurysm coiling was found to have a power of 0.45 with the remaining eight image categories demonstrating a power of greater than 0.98. Taken together, this suggests that our study is sufficiently powered to detect meaningful statistical differences across the vast major of image categories, ultimately allowing for adequate validation of the advanced generation strategy.

Our analysis reveals the fusiform aneurysm image showed a total score improvement from 22.40 (simple) to 35.00 (advanced), with a highly significant p-value of 7E-08. Similarly, the anterior cerebral artery image saw its total score increase from 17.20 to 27.40 ($p = 8E\text{-}08$), indicating a substantial enhancement in overall image quality.

The flow-diverting stent image also demonstrated marked improvement, with the total score rising from 27.20 to 35.80 ($p = 2E\text{-}08$), reflecting the advanced strategy's ability to generate clearer and more anatomically accurate representations. The saccular aneurysm image, another well-performing illustration, improved from 20.20 to 35.40 ($p = 2E\text{-}10$).

However, some images showed more modest improvements in their total scores. For instance, the aneurysm coiling image saw its total score increase from 18.00 to 21.80, but the p-value of 0.07 indicated that this improvement was not statistically significant. Similarly, the orbitozygomatic craniotomy image improved from 15.40 to 23.20, with a p-value of 3E-05, reflecting a statistically significant but less pronounced improvement compared to other illustrations.

Assessment of inter-rater reliability at this stage revealed good [24] consistency for both the simple generation strategy (ICC = 0.69, $p = 4.5E\text{-}12$, 95 % CI [0.55, 0.79]) and advanced generation strategy (ICC = 0.70, $p = 7.8E\text{-}13$, 95 % CI [0.57, 0.80]).

Overall, the results of Stage 2 demonstrated that the advanced generation strategy produced statistically significant improvements in most images' total scores, particularly for complex neurovascular structures like fusiform aneurysms and flow-diverting stents. While the advanced strategy led to notable enhancements across most categories, images like aneurysm coiling exhibited smaller gains, suggesting potential areas for further refinement.

3.3. Time and prompts required for advanced generation strategy

In addition to image quality, the advanced prompt generation strategy was evaluated based on two key metrics: the time required to generate each image, and the number of prompts used during the iteration phase (Supplemental Figure 1).

Across all image categories, the time and number of prompts required to generate an illustration varied based on the complexity of the neurovascular structure. The aneurysm clipping required the most time at 115 min while the orbitozygomatic craniotomy required the most prompts at 78. On the other end, the arteriovenous malformation required the least, with 45 min and 22 prompts. All other images, such as the fusiform aneurysm (62 min, 42 prompts) and aneurysm coiling (88 min, 53 prompts), fell between these two extremes. In general, time allotted and prompts required generally seemed to correlate. On average, generating an illustration took approximately 79 min and required 49 prompts.

These metrics underscore the iterative nature of the advanced generation strategy, demonstrating the balance between image complexity and refinement needs.

3.4. Stage 3: Validation of the advanced model

In the final validation phase, we evaluated the advanced generation strategy using a larger and more diverse group of participants, ranging from medical students to attending neurosurgeons. Their scores were weighted based on their level of training, allowing for a more detailed analysis of the image quality across different levels of expertise (Table 2).

Overall, saccular aneurysm images scored 33.47, with 85 % of respondents indicating they would use the image in their manuscripts without modification. The image achieved high marks across multiple criteria, including color (4.43), aesthetic quality (4.43), and accuracy (4.22). Similarly, the fusiform aneurysm image scored 32.50, with 77 % of evaluators approving the image as is. Notable scores included accuracy (4.38), location of structures (4.02), and color (4.08).

In contrast, the anterior cerebral artery image received lower scores overall, totaling 21.09, with only 8 % of respondents finding the image suitable for inclusion in its current form. Scores for accuracy (2.18),

Table 2
Advanced Generation Strategy Scores. Criterion scores represent the weighted average of scores across all participants. Total scores reflect the sum of all criteria. Percentages reflect the percent of participants that stated they would use the given image without modification. Note: scores are represented visually with dark green reflecting higher scores and white/light green reflecting lower scores.

Criteria	Saccular Aneurysm	Fusiform Aneurysm	Bifurcation Aneurysm	Anterior Cerebral Artery	Arteriovenous Malformation	Aneurysm Clipping	Aneurysm Coiling	Flow Diverting Stent	Orbitozygomatic Craniotomy
Accuracy	4.22	4.38	2.85	2.18	2.83	2.18	2.35	3.89	1.56
Location of Structures	3.97	4.02	3.28	2.20	2.75	2.58	2.29	3.66	1.68
Size / Scale	4.29	3.98	3.38	2.48	2.83	2.72	2.86	3.89	1.55
Color	4.43	4.08	3.57	3.62	3.14	3.12	3.23	3.88	2.74
Complexity	4.04	3.75	2.98	2.57	2.79	2.34	2.50	3.57	1.84
Educational Value	3.96	4.02	3.25	2.20	2.78	2.60	2.33	3.63	1.43
Relevance	4.13	4.26	3.32	2.49	3.22	3.11	2.88	3.63	1.54
Aesthetic Quality	4.43	4.02	3.32	3.36	2.82	2.72	2.86	4.03	2.37
Total Score	**33.47**	**32.50**	**25.94**	**21.09**	**23.16**	**21.36**	**21.28**	**30.16**	**14.70**
Would you include this image in your manuscript as is?	***85%***	***77%***	***38%***	***8%***	***31%***	***8%***	***8%***	***77%***	***15%***

educational value (2.20), and relevance (2.49) were particularly low, indicating that complex anatomical features like cerebral arteries presented challenges even for the advanced generation strategy.

Another area of relative weakness was aneurysm coiling, which scored 21.28 overall, with 8 % of respondents approving the image for manuscript use. Scores for accuracy (2.35) and size/scale (2.86) were lower than in other categories, again highlighting the limitations of the model when tasked with more intricate neurovascular procedures.

The flow-diverting stent image, however, received a more favorable response, with 77 % of respondents approving it for use and an overall score of 30.16. The image was particularly well-rated for accuracy (3.89), color (3.88), and complexity (3.57). This result suggests that the model excelled in generating images of neurovascular interventions that are relatively less anatomically complex but still require precision and clarity in visual representation.

Assessment of inter-rater reliability at this stage revealed excellent [24] consistency between evaluators across all levels (ICC = 0.88, p = 5.4E-61, 95 % CI [0.84, 0.92]). However, weighted analysis based on evaluator training level revealed some, more nuanced, insights. More experienced evaluators, such as attending neurosurgeons, tended to score images lower in categories such as accuracy and educational value, particularly for images like arteriovenous malformations (total score 23.16) and aneurysm clipping (21.36). In contrast, medical students and fellows often rated the images higher in categories like color and aesthetic quality, emphasizing the perceived educational utility of the illustrations.

4. Discussion

This study demonstrates the potential of AI-generated medical illustrations to alleviate the burden of producing such images, particularly in highly specialized fields like neurosurgery. With the growing demand for high-quality anatomical images for educational, clinical, and research purposes, the need for cost-effective and scalable solutions is pressing. While AI models offer a promising alternative, our findings highlight both the opportunities and limitations of these technologies in neurovascular surgery.

Our results show that the Gemini model outperformed DALL-E, Copilot, and Midjourney across most criteria, particularly in accuracy and educational value. The advanced generation strategy significantly improved image quality, especially for complex structures like fusiform aneurysms and flow-diverting stents. However, certain illustrations, such as the anterior cerebral artery and aneurysm coiling, consistently received lower scores, indicating that more intricate neurovascular anatomy remains challenging for AI models to depict accurately.

These findings align with previous studies evaluating AI-powered text-to-image generators for medical illustrations. Noel et al. (2024) reported similar limitations with DALL-E and other models, noting the omission of critical anatomical landmarks like foramina and suture lines in their illustrations [13]. Our study corroborates their conclusion that current AI models, while fast and cost-efficient, struggle with producing anatomically precise and clinically useful images of complex structures. This suggests that further training of AI models with anatomically correct datasets is essential to overcoming these limitations.

In terms of cost-effectiveness, AI-generated illustrations provide significant advantages, particularly in reducing the time and expense associated with hiring medical illustrators. As Amri and Hisan (2023) argue, the accessibility of AI-generated images can democratize the production of medical illustrations, particularly in resource-limited settings [1]. However, this benefit is counterbalanced by the need for human oversight to ensure the accuracy of the generated images, a limitation echoed by Lazarus et al. (2024), who emphasized that AI should not be a wholesale replacement for human illustrators [25].

One of the key insights from our study is why the advanced prompt generation strategy led to more accurate and higher-quality illustrations. Gemini's superior ability to interpret iterative prompts and refine images based on feedback suggests that AI models excel when guided through a structured and expert-driven process. This mirrors findings by Noel et al. (2024), who also observed that AI-generated illustrations improved with greater input and refinement [13]. However, even with these improvements, certain models struggled with more complex anatomical structures, like the anterior cerebral artery, highlighting that AI is not yet fully capable of handling the nuances of neurovascular anatomy.

Our study also offers insights into why the strategy failed in some cases. For example, the lower scores for anterior cerebral artery illustrations can be attributed to the inherent complexity of neurovascular anatomy, where subtle spatial relationships between vessels are critical. AI models, as noted by Lazarus et al. (2024), tend to oversimplify or miss key details in such contexts, creating a false sense of anatomical normalcy [25]. This reinforces the need for continuous refinement of both AI models and the prompts used to guide them.

One of the most critical limitations of this study is reproducibility. Even if future researchers replicate our exact prompt strategy, they may not generate the same images. This issue has been highlighted in several AI studies, including Lazarus et al. (2024), who emphasized the unpredictable nature of AI outputs due to the variability in training data and algorithmic processing [25]. While this is a significant challenge, our study serves as a general guide for others looking to develop AI-based medical illustration pipelines, providing insights into how to approach model selection, prompt design, and evaluation. Importantly, our study relied on a relatively modest sample size of evaluators, comprising all members of the neurovascular surgery service line at our institution. To ensure a broad perspective on the value and accuracy of image generation is captured, it is crucial that future studies expand the evaluator pool as utility of these images may differ based on the needs of the evaluator and their patient population.

Other limitations include the bias in AI training datasets, which may not fully capture the diversity of neurovascular anatomy. Amri and Hisan (2023) also noted that while AI models offer scalable solutions, they may propagate biases embedded in their training data, particularly when datasets are not representative of diverse anatomical features [1]. Additionally, the educational value of AI-generated illustrations, while promising, remains contingent on the ability of AI to depict subtle anatomical variations that are critical in neurosurgical contexts. This is a concern raised by Noel et al. (2024), who stressed that current AI models frequently omit important anatomical details [13].

To improve the utility of AI-generated illustrations in neurosurgery, future efforts should focus on refining training datasets to better capture the intricacies of neurovascular anatomy. Noel et al. (2024) suggest that expanding the scope of AI training datasets with anatomically correct images could enhance both accuracy and clinical relevance [13]. Additionally, advancements in AI models that prioritize detail over speed may address the current shortcomings in depicting complex structures like aneurysms and cerebral arteries.

Finally, the ethical implications of using AI-generated illustrations in clinical practice and education should not be overlooked. As Lazarus et al. (2024) point out, reliance on AI-generated images without proper validation could lead to oversights in critical learning environments [25]. Therefore, AI should be seen as a complementary tool to human illustrators, enhancing the speed and accessibility of medical illustrations while still requiring human expertise for final validation.

5. Conclusion

This study highlights the potential of AI-generated medical illustrations, particularly with the Gemini model, to alleviate the burden of developing high-quality medical illustrations in neurosurgery by providing cost-effective, scalable solutions. The advanced generation strategy led to significant improvements in image accuracy and educational value, though challenges remain in depicting complex neurovascular anatomy. Beyond demonstrating AI's capabilities, this publication establishes a systematic and rigorous framework for generating and evaluating AI-based medical illustrations, providing a foundation for future work in the field. As AI models continue to evolve, their integration into neurosurgical practice and education holds promise, but success will depend on addressing key limitations such as reproducibility, accuracy, and ethical considerations.

CRediT authorship contribution statement

Pierce Davis: Writing – review & editing, Visualization, Project administration, Investigation, Data curation, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Visish M. Srinivasan:** Writing – review & editing, Supervision, Conceptualization, Validation, Data curation. **Avi A. Gajjar:** Writing – review & editing, Investigation, Validation, Data curation. **Sneha Sai Mannam:** Writing – review & editing, Investigation, Validation, Data curation. **Sonia Ajmera:** Validation, Data curation, Writing – review & editing, Investigation. **Rashad Jabarkheel:** Validation, Data curation, Writing – review & editing, Investigation, Conceptualization. **Chethan Reddy:** Writing – review & editing, Investigation, Data curation, Validation, Formal analysis. **Alan Napole:** Writing – original draft, Validation, Project administration, Investigation, Data curation, Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Conceptualization. **Albert Q. Wu:** Writing – review & editing, Investigation, Validation, Data curation. **Sai Krishna Palepu:** Writing – review & editing, Project administration, Formal analysis, Software, Investigation, Data curation. **Redi Rahmani:** Validation, Data curation, Writing – review & editing, Investigation. **Sandeep Kandregula:** Validation, Data curation, Writing – review & editing, Investigation. **Jan-Karl Burkhardt:** Validation, Investigation, Writing – review & editing, Supervision, Data curation. **Joshua S. Catapano:** Writing – review & editing, Supervision, Data curation, Validation, Investigation. **Kyle W. Scott:** Validation, Data curation, Writing – review & editing, Investigation. **Hasan Ahmad:** Validation, Data curation, Writing – review & editing, Investigation. **Oleg Shekhtman:** Validation, Data curation, Writing – review & editing, Investigation. **Georgios Sioutas:** Validation, Data curation, Writing – review & editing, Investigation.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.clineuro.2025.109039.

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